

The Pigeon Hole Principle Worksheet

1. A set of cards contains red, green and blue cards. How many cards must Jason choose at random to ensure that there are at least 3 cards with the same colour?
2. Henry chooses 3 random distinct positive integers, prove that either at least two of the numbers are odd or at least two of the numbers are even.
3. I have 10 DotPoint Physics, 10 Cambridge 3U, and 10 Maths in Focus. At least how many books must I take out in random to ensure that I have 3 copies of Cambridge 3U?
4. There are 500 people at a concert. Explain why at least 2 of them have the same birthday.
5. How many numbers does John need to pick from 1 to 8 to ensure that at least a pair will add up to 9?
6. Study shows that year 11 students are good at one and only one of the following 3 topics: calculus, probability, or geometry. How many random students must I talk to ensure that at least 3 of them are good at the same topic?
7. In my drawer, I have 5 gloves for left hand and 7 for right hand. At least how many gloves must I take out at random to ensure I have a glove for each hand?
8. Kurt needs to split 174 students into 7 classes. Find the minimum number of students in at least one of these classes.
9. In a survey of 379 students, there were at least 28 students who hated the same maths topic. How many different topics were surveyed?
10. How many students must be split into 8 groups so that there are at least 29 students in at least one group?
11. The Cambridge textbook has 2 versions, 3 Unit and 4 Unit. Each version has 3 editions, regular, premium and deluxe. How many random Cambridge textbooks must I buy so that at least two of them are the same version AND edition?
12. At Bamboo Express, a Chinese restaurant, customers can order rice with 1 dish, out of 12 choices of dishes. How many customers came in today so that at least 22 of them chose the same dish?
13. Show that from any 14 odd integers chosen from $\{1, 3, 5, \dots, 49, 51\}$, at least a pair will add up to 52.
14. Kurt now has 4 new detention rooms (so 5 in total). Please assume that each room is sufficiently large for the purpose of this worksheet. He has sent all 431 year 11 students to detention. Find the minimum number of students in at least one of the detention rooms.
15. Again, Kurt now has 4 new detention rooms plus the original one. All 431 year 11 students did poorly in the test so they all got detention. This time the 4 new rooms are empty and the original room already has 24 year 12 students in it. Find the minimum number of students in at least one of the detention rooms.

16. After 2 detentions, all of the year 11 students (431 in total) still didn't pass Kurt's test and thus got detention. Again, the 4 new rooms are empty and the original room has x number of year 12 students in it. If the minimum number of students in at least one of the detention rooms is also x , find x .
17. Unfortunately (or fortunately?), Kurt's detention rooms are now filled with year 11 and 12 students, so he asks the y number of year 10 students to each choose 1 out of 50 worksheets to complete at home. Supposed that 25% of the students have chosen one of worksheet 1 to 10 (remaining 75% have each chosen one of 11 to 50), what is y if we're certain at least three students have chosen to do the same worksheet? (Note: 25% of y is rounded up, e.g. 25% of 145 students is 37 students)
18. Kurt also offers to play a game, if you win, you can skip detention. He has a set of 50 cards numbered 1 to 50. You throw a die to determine how many cards you can pick at random. (e.g. if you throw 3, you pick 3 random cards) You win if the difference between at least one pair of cards is less than or equal to 12. What number(s) from the die must you throw to ensure that you will definitely win?
19. You are still offered the same game, except this time, you win if the difference between at least one pair of cards is less than or equal to z . You realised that throwing 3 or higher will guarantee your win. Find the minimum value of z .
20. A chessboard has 8 columns and 8 rows. Rooks will attack each other if placed in the same column or row. What's the greatest number of rooks I can place on the board before any two of them will attack?
21. There are 51 points randomly scattered in a 5m by 5m square. Show that there is a 1m by 1m square containing at least 3 points.
22. Suppose that for newly enrolled high school students, the student number now uses the following rule: The first letter of their last name plus their birthday in DDMM. For Mary Lee born on 9th May, her number is L0905. The rule is in effect from 2019. Suppose that each year, 4000 students enrol in schools (so 4000 in 2019, 4000 in 2020 etc.). In which year are we certain that there is a repetition in student number?

Answer:

Q1: 7	Q3: 23	Q5: 5	Q6: 7	Q7: 8	Q8: 25	Q9: 14	Q10: 225
Q11: 7	Q12: 253	Q14: 87	Q15: 91	Q16: 108	Q17: 81	Q18: 5 or 6	Q19: 24
Q20: 8	Q22: 2021						